

CHRIS TALBOT | CV

Ph.D. Student in Computational Biology, Cornell University
National Science Foundation Graduate Research Fellow

Status: Ph.D. Student in Computational Biology at Cornell University
Fields: Population Genetics, Computational Biology, Evolutionary Ecology
Interests: Global change, rapid evolution, stochastic modeling, conservation
Techs: C, C++, Python, TypeScript, CUDA, Cerebras, SLiM

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Education

Ph.D. Computational Biology - Cornell University 2025-Present

- Advisor: Dr. Philipp Messer
- Coursework: High-performance and parallel computing, population genetics

B.S. Ecology, Evolution, and Biodiversity - University of Michigan (UMich) - GPA 3.74 2024

- Graduated with **Highest Honors** under the supervision of Dr. Marjorie Weber
- Honors Thesis: Patterns of floral color in communities of common Northeast American wildflowers
- Math & CS Coursework: Statistics & probability, programming & data structures, real analysis, discrete math
- Biology Coursework: genetics, macroevolution, woody plants, ethnobotany, field ecology

Associate of General Studies - Grand Rapids Community College (GRCC) - GPA 3.9 2020

- Computer Science & Mathematics Concentration
- Coursework: C++ programming, database development, web application programming, information security
- Invited member of Phi Theta Kappa Honors Society

Research

Doctoral Researcher-Messer Lab - Cornell University-Computational Biology 2025-Present

Research Programmer-Bradburd Lab - UMich-Ecology & Evolutionary Biology (EEB) 2024-2025

Undergraduate Researcher, Lab Technician-Weber Lab - UMich-EEB 2022-2025

Undergraduate Researcher-Márquez Lab - UMich-EEB 2021-2022

Publications

Published:

2026 – Talbot, C. A., Bradburd, G.S. ARGscape: A modular, interactive tool for manipulation of spatiotemporal ancestral recombination graphs. arXiv:2510.07255 [q-bio.PE]. **Published in *Bioinformatics*.**

2026 – Glos, R., Soule, A. R., ..., Talbot, C. A., ..., Weber, M. G. Micromorphology and Chemistry of Mite Domatia: Mutualistic Trait Convergence and Variation across Eight Sympatric North American Species. **Published in *Symbiosis*.**

2025 – Graham, C. D. K., ..., Talbot, C. A., Yerks, A. L., Weber, M. G. Mite domatia and associated mite density in a North American Eastern Deciduous Forest in Michigan. **Published in *Ecology and Evolution*.**

Pre-prints/In Review:

2026 – Talbot, C. A., Weber, M. G. Opposing seasonal dynamics in phylogenetic and flower color diversity of co-flowering wildflower assemblages. bioRxiv 2025.10.21.683724. **In Revision at *Ecosphere*.**

In Prep:

2026 – Talbot, C. A., Bradburd, G. S. Optimizing temporal sampling strategies for spatial population genetic inference using ancestral recombination graphs. **In Preparation for *Genetics*.**

2026 – Talbot, C. A., Kern, A. D. slim-vscode-tools: A comprehensive language server and development environment for SLiM forward-time simulations. **In Preparation for *Bioinformatics*.**

Funding, Honors, & Awards

Total: \$174,180

Graduate Research Fellowship - National Science Foundation (\$159,000)	2025-2030
Graduate School Dean's Scholar Fellowship - Cornell University (\$2,000)	2025
Chancellor's Fellowship - University of California–Berkeley (Declined)	Offered 2025
Eugene Cota-Robles Fellowship - University of California–Davis (Declined)	Offered 2025
Eugene Cota-Robles Fellowship - University of California–Santa Cruz (Declined)	Offered 2025
Biological Station Student Fellowship - UMich Biological Station (\$1,680)	2023
Student Research Grant - UMich UROP (\$5,000)	2022-2023
Travel Grant - SACNAS (\$2,500)	2022
Biomedical & Life Sciences Summer Fellowship - UMich UROP (\$4,000)	2022

Invited Talks & Workshops

ARGEvol Conference - Porto, Portugal (Invited Talk)	2025
ARGEvol Spatial ARG Workshop - Porto, Portugal (Workshop Contributor)	2025
tskit-dev Seminar - Virtual (Invited Talk)	2025

Contributed Talks

Third Joint Congress on Evolutionary Biology - Montreal, QC (Talk)	2024
Botany 2024 - Grand Rapids, MI (Lightning Talk)	2024
UMich Ecology & Evolutionary Biology - Ann Arbor, MI (Honors Thesis Defense)	2024

Posters

Third Joint Congress on Evolutionary Biology - Montreal, QC	2024
Botany 2024 - Grand Rapids, MI	2024

Software Development

Open-Source Contributor - "SLiM" software, Messer Lab, Cornell University	2025-Present
· Contribute to maintenance of software for simulating genetics, evolution, and ecology	
Open-Source Lead - "ARGscape" software	2025-Present
· Design and maintain software for simulating, analyzing, and visualizing genomic data	
Open-Source Lead - "slim-vscode-tools" software	2025-Present
· Design and maintain software for streamlining SLiM programming through a language server and IDE extension	

Undergraduate Mentoring

McNair Scholars Graduate Coordinator - Cornell University	2026-Present
· Paid position involving teaching courses, direct mentorship, and program administration	
Peer GRFP Mentor - Cornell University	2025
· Paid position involving direct mentorship and peer review for GRFP applicants	
Transfer Student Peer Mentor - UMich-LSA Transfer Bridges Program	2022-2023
· Paid position mentoring prospective transfer students and reviewing application materials	

Undergraduate Teaching

Co-Instructor - Cornell McNair Scholars Summer Writing Course	2026
Course Assistant - "Insights From Trees," University of Michigan Biological Station	2024, 2025
Professional Peer Tutor - GRCC Drop-in Tutoring Labs - Mathematics, Language Arts	2019-2020

K-12 Teaching

Volunteer Educator - Discover Cayuga Lake-Trout in the Classroom	2025-Present
Volunteer Educator - Skype a Scientist	2025-Present
Docent (Volunteer Educator) - Matthaei Botanical Gardens & Nichols Arboretum	2023-2025
Test Writer & Proctor, Forestry & Botany (Volunteer) - UMich Science Olympiad	2023-2025
Director of Percussion - Lake Orion Community Schools, Lake Orion, MI	2022-2023
Director of Percussion - Spring Lake Public Schools, Spring Lake, MI	2019-2022
Freelance Percussion Instructor & Composer - 8 school districts across Michigan	2018-2022

Professional & Service

Ad-hoc Peer Reviewer - <i>Evolution Letters</i>	2026
Co-Founder, President, and Volunteer - Students for Public Power @UM	2023-2025

Professional Memberships

American Society of Naturalists - Student Member	2024-Present
Society for the Study of Evolution - Student Member	2022-Present
Botanical Society of America - Student Member	2024-2025

Languages

German - B2, Intermediate	
Nishnaabemwin (Eastern Ojibwe) - A1, Beginner	

Certifications

Certified Michigan Naturalist - Michigan State University Extension	2024
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Extracurriculars

Percussionist - Cornell Percussion Ensemble	2025-2026
Percussionist - Straits of Mackinac Concert Band, Mackinaw City, MI	2024-2025
Founding Member - UMich Ecological Restoration Club	2023-2025
Vice President - GRCC Programmers	2019-2020